EnterpriseGuard

ENTERPRISE DATA BREACH PREVENTION SYSTEM

Domain: Database Management Systems

Focus: Data Analysis & Security

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PROBLEM STATEMENT

Enterprises handle large volumes of sensitive data accessed by multiple employees across various systems, making them vulnerable to insider threats and unauthorized access. Existing logging mechanisms lack effective analysis and early detection of suspicious behavior. There is a need for a centralized DBMS-based solution that records access activities, analyzes risk patterns, and proactively generates alerts to prevent potential data breaches.

Tech Stack
Databse: MySQL
Frontend: HTML, CSS, JavaScript
Backend: Python (Flask)
Core Concepts applied: ER Modelling, Normalisation, PK-FK Relationships

PROJECT OBJECTIVES

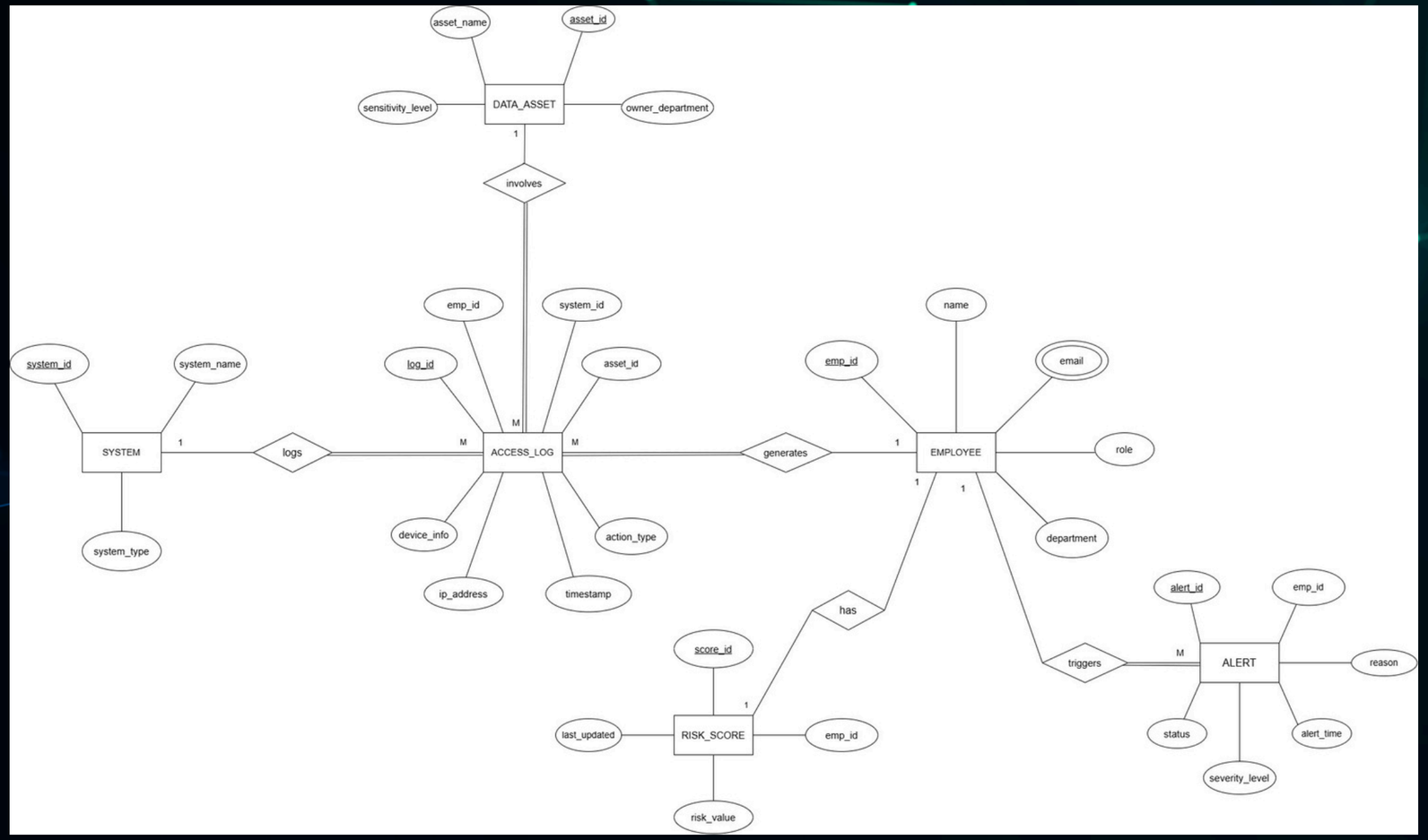
1. Store enterprise systems, employees, and data assets
2. Log every access activity in a centralised database
3. Generate alerts for suspicious activities
4. Analyse access behaviour using DBMS concepts
5. Assign risk scores to employees

SYSTEM OVERVIEW

1. Employees access enterprise systems
2. Access to sensitive data assets is recorded
3. Each access generates an access log entry
4. Risk score is calculated based on access behaviour
5. Alerts are triggered when risk exceeds threshold

ER DIAGRAM OVERVIEW

1. This system is designed using an ER model
2. Entities represent real-world enterprise objects.
3. Relationships define interactions between entities.
4. Primary and Foreign keys maintain data integrity.



ENTITY DESCRIPTION

Entities in the system:

EMPLOYEE

SYSTEM

DATA_ASSET

ACCESS_LOG

RISK_SCORE

ALERT

Each entity stores structured information required for tracking, analysis, and security monitoring.

RELATIONSHIPS & CARDINALITY

Cardinality ensures real-world accuracy.

1. One EMPLOYEE generates many ACCESS_LOG records
2. Each EMPLOYEE has exactly one RISK_SCORE
3. One SYSTEM logs many ACCESS activities
4. One DATA_ASSET can be involved in many ACCESS_LOGS
5. One EMPLOYEE can trigger multiple ALERTS

DATA BREACH DETECTION LOGIC

1. Access logs are analysed to detect abnormal behaviour

2. Risk score increases when:
   - Sensitive data is accessed frequently
   - Access occurs outside normal hours
   - Repeated suspicious actions are detected

3. Alerts are generated when risk crosses a threshold

CONCLUSION AND FUTURE SCOPE

CONCLUSION

- The ER Diagram effectively models enterprise data security
- DBMS concepts ensure data integrity and analysis
- System helps in early detection of insider threats.

DEFINING SCOPE

- Machine Learning based anomaly detection
- Real-time alert notifications.
- Integration with enterprise security system.